

PROBLEM STATEMENT #19 | Healthcare & Telemedicine

Personalized Meal & Diet Subscription Manager

1. Problem Context & Overview

A clinical dietary subscription service that generates weekly menus based on patient metabolic constraints (allergies, diabetic limits, calorie goals) and tracks daily delivery fulfillment.

Target Stakeholders / Actors: *Subscriber, Dietitian*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall generate weekly meal plans strictly adhering to user macro-nutrient constraints and allergen exclusion lists.

Sample Acceptance Criteria: Pass: Generated menu contains zero excluded allergen ingredients; Fail: Peanuts included for peanut-allergic profile.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The subscription engine must support schedule modifications (pause, skip, reschedule) up to 12 hours before dispatch.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.